

MedGemma recall benchmark: MDACE billing extraction

How much of the billed evidence a 4B open model recovers from a clinical note.

EXECUTION PLAN
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WHAT IS BEING MEASURED

MedGemma reads a note and extracts medical findings. **It does not produce codes.** Each extracted term then goes to the SNOMED lookup, which returns the adjacent ICD code. Asking a model for the code directly compounds two hard problems into one and produces too many errors.

Extraction is therefore the ceiling on the whole pipeline: the lookup can never recover a code whose term was never extracted. This benchmark measures where that ceiling sits for **MedGemma-4B, zero-shot**. There is no target to hit. It reports the best the model achieves, and Ehtesham Bhai decides whether that clears the bar.

DATA

8-07-mdace-ner-eval_sample_100-LOCAL.jsonl, and nothing else. Every note in the file is used.

24	100	91
NOTES	ANNOTATION ROWS	DISTINCT CODES

WHAT COUNTS AS A CORRECT EXTRACTION

Three columns describe each billed code. **Matching any one of them counts.**

Evidence text	what the note says (<i>depression</i>)
Code description	ICD catalogue wording (<i>Major depressive disorder, single episode, unspecified</i>)
SNOMED terms	clinical synonyms for the same concept

That is a median of **4 accepted phrasings per code**. Recall is reported both per row (100) and per distinct code (91).

HOW A MATCH IS DECIDED

Matching is approximate, not exact. Each level contains the one above it, and **recall is reported at every level**, so the cost of each degree of looseness is visible rather than hidden inside one threshold.

LEVEL	RULE	CATCHES
L1 exact	identical after normalising	baseline
L2 contains	one is a whole-word subset of the other	back pain in chronic back pain
L3 similar	word overlap and character similarity	typos, word order
L4 embeddings	biomedical sentence model, cosine	HTN to hypertension
L5 LLM judge	a model decides whether two phrases mean the same	deferred, see below

L4 is the level that earns its keep. Abbreviations are the largest single source of missed matches, and no amount of string similarity reaches them: *CHF* and *congestive heart failure* share almost no characters. Only semantic matching closes that gap.

The model is also asked for two things per finding: the phrase as written in the note, and the standard clinical name for it. Either may match, and the standard name is what the lookup would receive in production anyway.

Scope. Extraction only. ICD code prediction is a separate experiment and is not measured here.

WHAT IS REPORTED

Recall	headline. Of the billed codes, how many were found
Precision, F1	secondary; false positives are watched, not ignored
Terms per note	the volume the recall was bought with
Not-in-note rate	hallucination: terms produced that are not in the text

Every figure is given at every matching level.

THE NUMBER THAT MAKES RECALL MEANINGFUL

With loose matching and no volume control, a model that lists every phrase in the note scores near 1.00. Recall alone therefore cannot rank two models: a smaller, more verbose one would appear to beat a larger, more disciplined one.

So recall is always quoted **with the terms-per-note it was bought with**: “0.72 at 20 terms per note”, never a bare 0.72. That is what keeps the benchmark usable when a larger model is measured against it later.

WHAT LIMITS THE NUMBER

SNOMED coverage in the file is partial. It carries 24,333 concept terms where the count field reports 270,287 in total, so roughly 9% are present, and procedure codes (CPT and ICD-10-PCS) carry none. Recall measured against that column is therefore a floor: some synonyms that would have matched are simply not there to match.

Approximate matching can be wrong in both directions. *Acute* and *chronic renal failure* differ by one word and are clinical opposites; a negated mention reads as a positive one. That is what the deferred audit exists to check.

DEFERRED: THE LLM JUDGE

Recall rises at every level *by construction*, so L4 will always look best. The honest question is which level is still counting genuine matches, and the way to settle it is to have a model judge a sample of the pairs each level newly accepted.

It is slow and non-deterministic, so it is **not run for now**. It stays as the audit step, to be added when a particular level's number is challenged or when the result sits close to a decision.

STARTING POINT

Scored the strict way (exact matching against the note wording alone) MedGemma returned **recall 0.53** on an initial run, against the ≈0.5 Ehtesham Bhai observed independently. That is the floor this plan is designed to lift, and every change above widens what counts as correct without touching the model itself.

MedGemma-4B-it · 4-bit · greedy · zero-shot